UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 015 · Applications of LDEs: simple harmonic oscillation · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S015·11	Unit 2 deck 015 (“Applications of DE – SHM”), slide 11 (rendered .pptx)
T015 00:11:48	Unit 2 AV-summary transcript “Application problems on linear differential equations – Simple harmonic oscillation” (16.3 min), segment start
CW U2 P37	Unit 2 class-work problem 37

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1. Figure G1.1 · The spring in three positions (slide 015·5)
2. Figure G2.1 · Motion of the 12 lb weight: $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$
3. Figure G2.2 · Displacement and velocity of the 8 lb weight
4. Figure G3.1 · Damped motion for $\lambda = 0.1, 0.3, 1$ and 2 $(n = 1, x_0 = 1)$

Slide 015·5 has the only diagram (the spring in three positions), redrawn in G1.1; every other figure is added for clarity. Colours follow the Okabe–Ito order: blue = displacement $x(t)$, vermilion = velocity or a second case, green = a third case; grey dashed lines are references (amplitude $\pm A$, envelopes, the period). Every plotted curve is the sympy-verified solution of its worked example. Positive x is downward, as in the deck.

1 The spring–mass model and simple harmonic motion

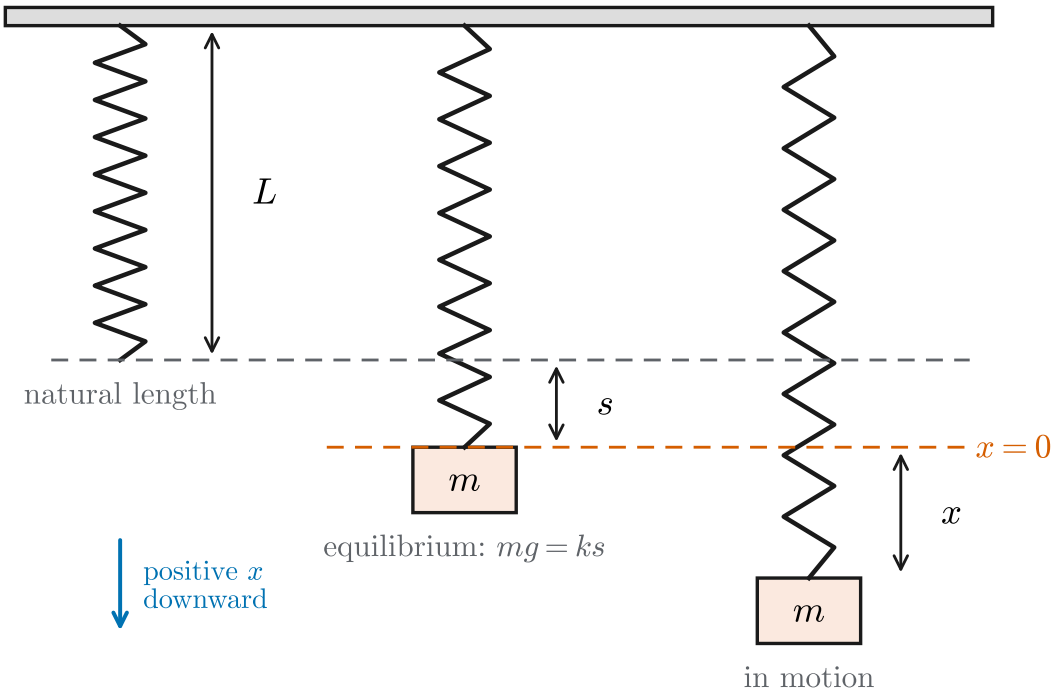


Figure G1.1. The spring in three positions (slide 015·5). ESA ★

S015·5 · T015 00:01:51

How to read it. Left: the natural length L. Middle: with the mass at rest the spring is stretched by s, where $mg = ks$; this is $x = 0$. Right: in motion the mass is x below equilibrium and the total stretch is $s + x$. Label L, s, x and $mg = ks$.

2 Solving spring problems

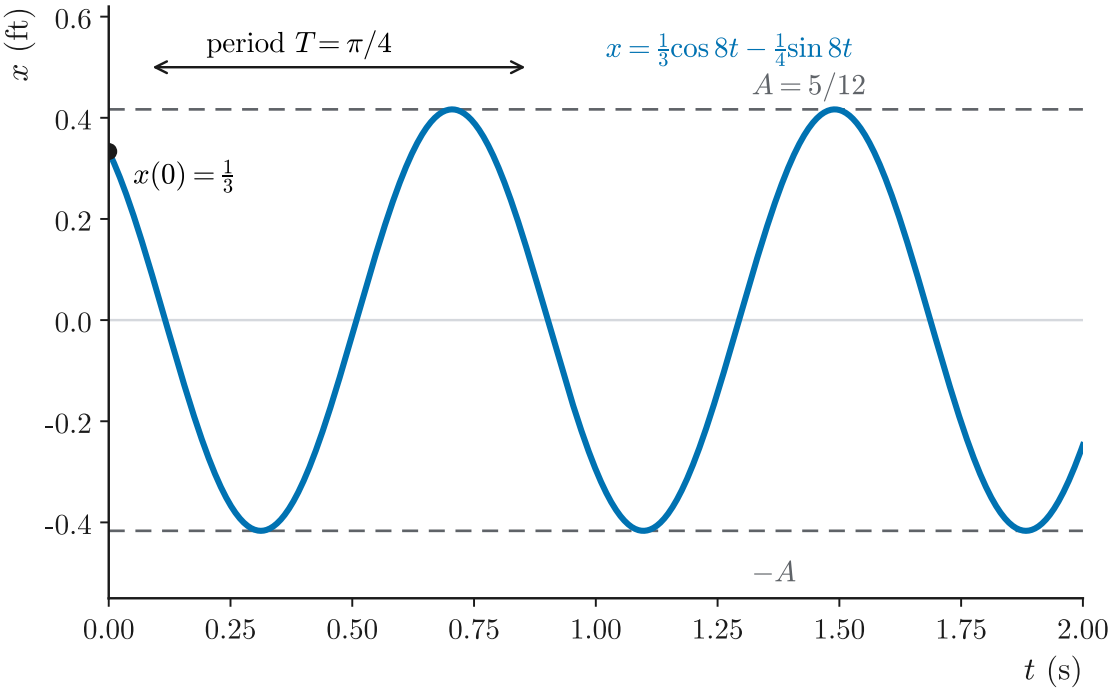


Figure G2.1. Motion of the 12 lb weight: $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$. ESA ★★

ADDED FOR CLARITY S015·14

How to read it. The mass starts $\frac{1}{3}$ ft below equilibrium (black dot) moving upward, so the curve starts downward-sloping on the plot. It swings between $\pm\frac{5}{12}$ ft (grey dashed) with period $\frac{\pi}{4} \approx 0.785$ s; the first crossing of 0 is at 0.116 s (W2.2). Positive x is below equilibrium.

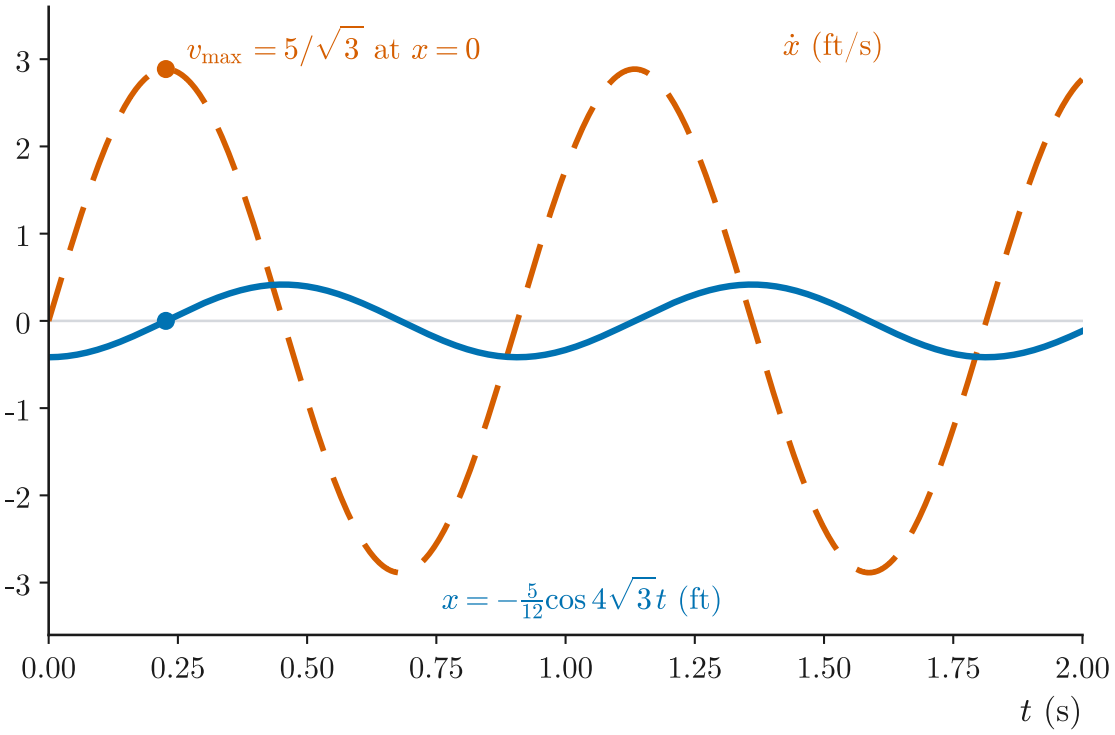


Figure G2.2. Displacement and velocity of the 8 lb weight. ESA ★★

ADDED FOR CLARITY S015·21 · S015·22

How to read it. Blue: $x = -\frac{5}{12}\cos 4\sqrt{3}t$ (ft). Vermilion dashed: $\dot{x} = \frac{5\sqrt{3}}{3}\sin 4\sqrt{3}t$ (ft/s). The velocity is largest, $\frac{5}{\sqrt{3}} \approx 2.89$ ft/s, exactly where $x = 0$, and zero at the turning points $x = \pm\frac{5}{12}$.

3 The damped oscillator

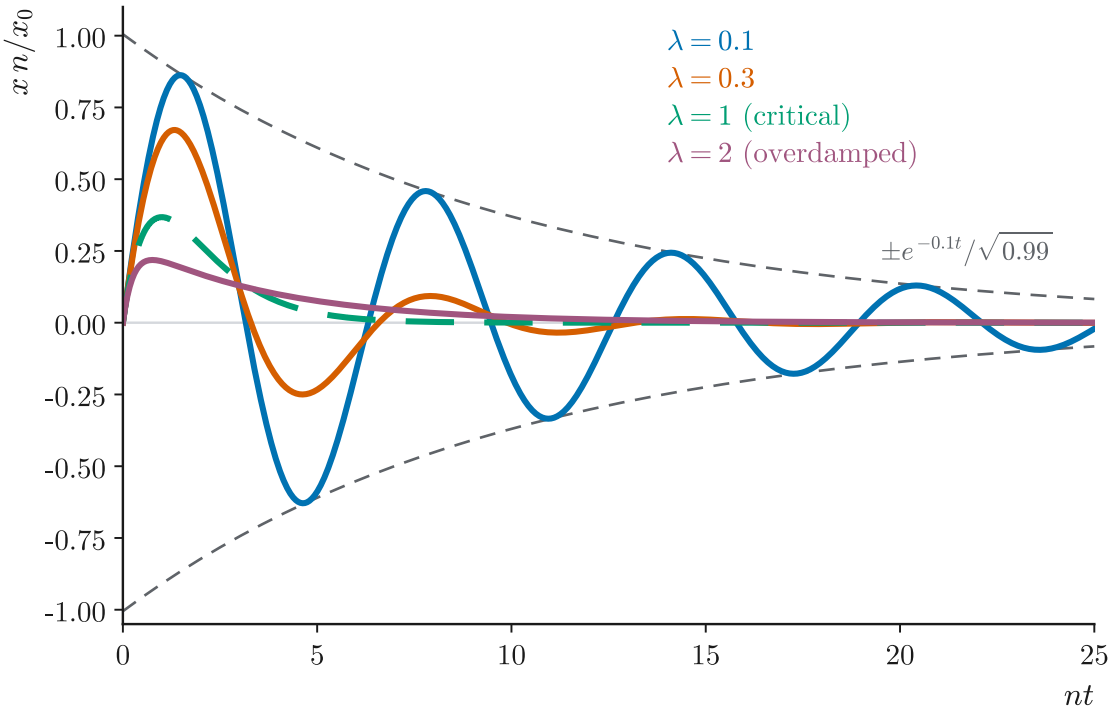


Figure G3.1. Damped motion for $\lambda = 0.1$, 0.3, 1 and 2 ($n = 1$, $x_0 = 1$). ESA ★

ADDED FOR CLARITY S015·16

How to read it. Underdamped curves ($\lambda = 0.1$ blue, 0.3 vermilion) oscillate inside their envelopes $\pm\frac{e^{-\lambda t}}{\sqrt{1-\lambda^2}}$ (grey dashed for $\lambda = 0.1$). Critical ($\lambda = 1$, green) and overdamped ($\lambda = 2$, purple) rise once and creep back without crossing zero.